



Low noise, self-referenced all polarization maintaining Ytterbium fiber laser frequency comb

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Abstract: We report the implementation of a self-referenced optical frequency comb generated by a passively mode-locked all polarization maintaining (PM) Yb fiber laser based on a nonlinear amplifying loop mirror (NALM). After spectral broadening the optical spectrum spans from 650 nm to 1400 nm, allowing for the generation of an optical octave and carrier envelope offset frequency (f_{ceo}) stabilization through a conventional f-2f interferometer. We demonstrate for the first time the stabilization of the f_{ceo} of such a PM Yb system with an in-loop fractional frequency stability scaled to an optical frequency of low 10^{-19} at 1 second averaging time, offering a great potential for applications in optical atomic clock metrology.

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1. Introduction

The most stable (short-term and long-term) and accurate electromagnetic radiation to date occurs in the optical domain [1–3], making optical radiation a critical tool for the validation of fundamental physics concepts [4–8], the redefinition of the second [9, 10] and the implementation of an optical timescale [11]. An optical frequency comb is generally required to serve as a clockwork that transfers the stability and accuracy of an optical reference (mostly an optical clock system) to other wavelengths of interest. As a prerequisite, the residual stability of the frequency comb has to be at least one order of magnitude better than that of the optical reference to ensure transfer without degradation. Single-port Er fiber-based frequency comb configurations have been experimentally demonstrated to realize very low residual instability of low 10⁻¹⁸ at 1 second averaging time for optical clock comparison and stability transfer [12–14]. However, due to the limited spectral coverage of Er combs (usually ~1 – 2 μm), a frequency doubler has to be installed in experiments involving the visible, where most of the high-precision optical spectroscopy applications occur. This doubling stage can jeopardize the fidelity of the stability transfer and comparison and adds additional layers of complexity. A way to overcome this restriction is the use of an Yb fiber-based comb system.

An Yb fiber oscillator operates in the 1.0 – 1.1 μm spectral region. After spectral broadening, coverage of both the visible and infra-red regions can be obtained [15]. Consequently, both visible-visible and visible-infrared stability transfer and comparison are permitted with all-common optical paths. Optical frequency comb systems based on non-

polarization maintaining Yb fiber laser oscillators have been experimentally demonstrated, with fractional instability below 10^{-17} at 1 second averaging time in-loop [16–20]. However, the intrinsic stability of these systems is compromised due to the polarization-sensitive nature of the laser oscillator. In contrast, all-polarization maintaining systems offer a much improved robustness as the laser oscillator is less vulnerable in the presence of environmental perturbations. An all-PM Yb fiber laser oscillator was reported recently in [21], based on a nonlinear amplifying loop mirror (NALM) [22], however without any comb stabilization.

In this paper, we present the realization of an optical frequency comb generated by an all-PM Yb fiber laser based on a nonlinear amplifying loop mirror. The laser oscillator has a mode-locked spectrum spanning $\sim 1015 - 1070$ nm with a relative intensity noise (RIN) of 0.18% integrated from 1 MHz to 1 Hz. After supercontinuum generation, an optical spectrum spanning from $\sim 650 - 1400$ nm is obtained. The detected carrier envelope offset frequency (f_{ceo}) has a signal to noise ratio (SNR) of 40 dB in a 100 kHz resolution with record ~ 7.5 kHz free-running linewidth. Stabilization of the f_{ceo} is demonstrated with a phase noise power spectral density of -90 dBc/Hz at 1 Hz Fourier frequency and a fractional frequency stability scaled to an optical frequency of 4×10^{-19} at 1 second averaging time. These low noise-level results are a good indication that all-PM Yb fiber frequency combs have a great potential in applications for precision optical atomic clock metrology as well as for timing distribution with attosecond precision.

2. All-PM Yb NALM frequency comb

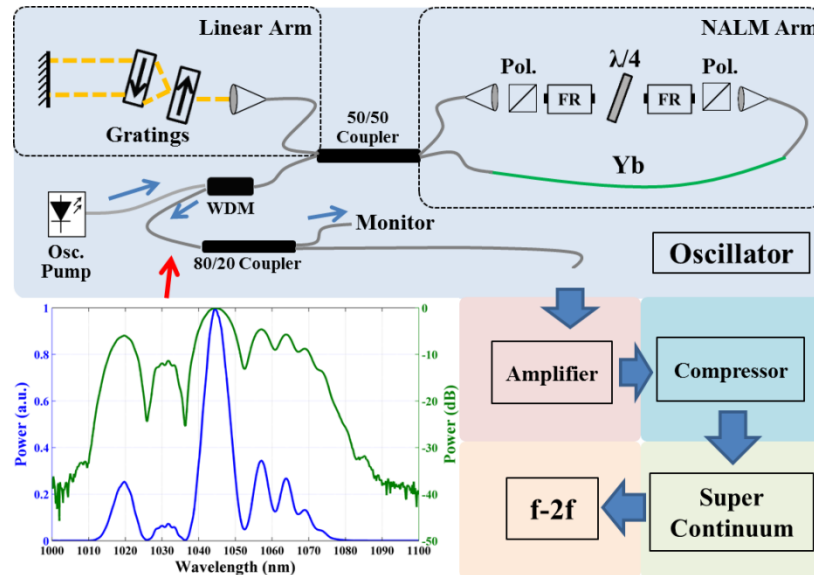


Fig. 1. Optical system schematic of the all-PM Yb NALM frequency comb. Inset: measured laser oscillator output spectrum in linear (blue) and log (green) scales. See text for details. WDM: wavelength division multiplexer; Pol.: polarizer; Osc.: oscillator; FR: Faraday rotator.

The complete optical portion of our all-PM Yb NALM frequency comb is depicted in Fig. 1. The design of the laser oscillator is essentially a derivative of our previously published Er system [23]. A phase bias module, consisting of a pair of polarizers, a pair of Faraday rotators and a quarter waveplate, is installed in the loop for the optimization of the mode-locking threshold and to ensure self-starting operation. A pair of gratings with 1000 lines per mm density and ~ 4.5 mm separation is introduced into the linear arm of the NALM for dispersion compensation [24]. The total loop fiber length is ~ 58 cm with ~ 28 cm high-doped PM Yb gain fiber, and that of the linear arm is ~ 20 cm. After a fiber coupler, 80% output of the laser oscillator is fed into the fiber amplifier while the remaining part serves as a monitor. The fiber

amplifier consists of a pre-amplifier with ~ 48 cm Yb-doped gain fiber and a power-amplifier with ~ 2.6 meters of Yb double-cladding fiber. The total fiber length of the amplifier is ~ 4 meters. Subsequently, the amplified optical pulses enter a free-space compressor with a pair of gratings (1000 lines per mm) separated by ~ 8.5 mm. No effort was made for third order dispersion compensation in the oscillator and amplifier. The compressed ultra-short pulses are coupled into a ~ 21 cm long photonic crystal fiber for super-continuum generation and f-2f generation.

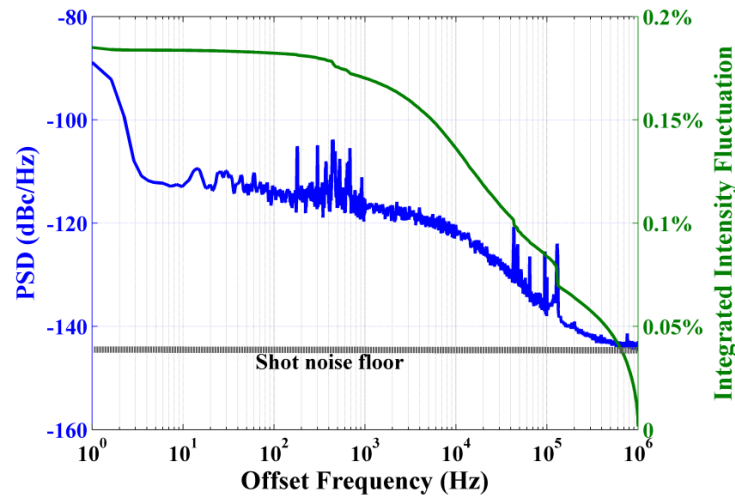


Fig. 2. Oscillator relative intensity noise measurements. Blue: power spectral density. Green: integrated relative intensity fluctuation. Shot noise floor at ~ 143 dBc/Hz.

When the laser oscillator is mode-locked the output power is measured to be 6.3 mW. The corresponding optical spectrum with ~ 55 nm 10-dB spectral width is plotted in the inset of Fig. 1 (linear scale in blue, log scale in green). The repetition rate is measured to be ~ 131 MHz. The power spectral density (PSD) of the laser oscillator relative intensity noise, measured by directly photodetecting the laser output, is plotted in Fig. 2. The relative intensity fluctuation, indicated by the square root of the integral of the PSD from 1 MHz to 1 Hz, is approximately 0.18%.

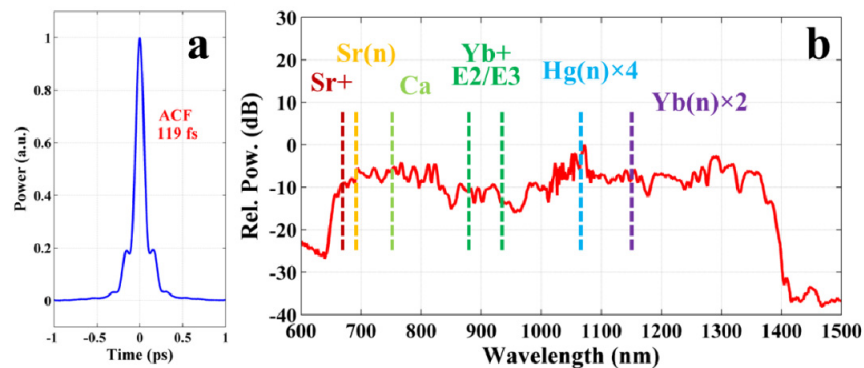


Fig. 3. (a) Autocorrelation trace of the amplified pulse after the compressor. (b) Measured spectrum of the octave-spanning super-continuum. Colored dash lines indicate approximate clock transitions of various optical atomic clocks.

With ~ 200 mW pre-amplifier pump power and ~ 4 W power-amplifier pump power, the measured optical power at the input of the PCF (NKT NL-1050-zero-2) is ~ 1.85 W. The

compressed pulse width, obtained by an autocorrelator, is approximately 80 fs, as shown in Fig. 3(a). Under a coupling efficiency of $\sim 60\%$, an octave-spanning super-continuum spectrum is generated, as shown in Fig. 3(b). As illustrated by the dashed lines in Fig. 3(b), the optical spectrum covers all the relevant clock transition candidates for the redefinition of the second. The comparison of optical clocks or stability transfer of ultra-stable lasers across the visible and infra-red can be realized without the need of a doubling stage and hence be used in a full common optical path setting. A continuum covering the $1.55\ \mu\text{m}$ spectral region can be expected by implementing additional third-order dispersion compensation or the implementation of optimized photonic crystal fiber [15].

By using a standard $f\text{-}2f$ interferometer [18], a free-running f_{ceo} with approximately 40 dB SNR in a 100 kHz bandwidth is obtained, as shown in Fig. 4(a). This SNR is sufficient for self-referenced phase locking. The free running f_{ceo} linewidth is measured by directly feeding the f_{ceo} signal to a digital phase analyzer referenced to an oven-controlled crystal oscillator (OCXO). The obtained power spectral density is plotted in Fig. 4(b). The linewidth is retrieved to be $\sim 7.5\ \text{kHz}$ at $1\ \text{rad}^2$ of the integrated PSD [25]. This value is in fact lower than achieved with other non-PM Yb fiber combs (though no exact definition of free running f_{ceo} linewidth was provided in previous papers) [16–18]. Moreover, the narrow f_{ceo} linewidth indicates that the timing jitter of the present Yb laser can be below 100 as [26], which makes it very attractive for precision timing distribution [27].

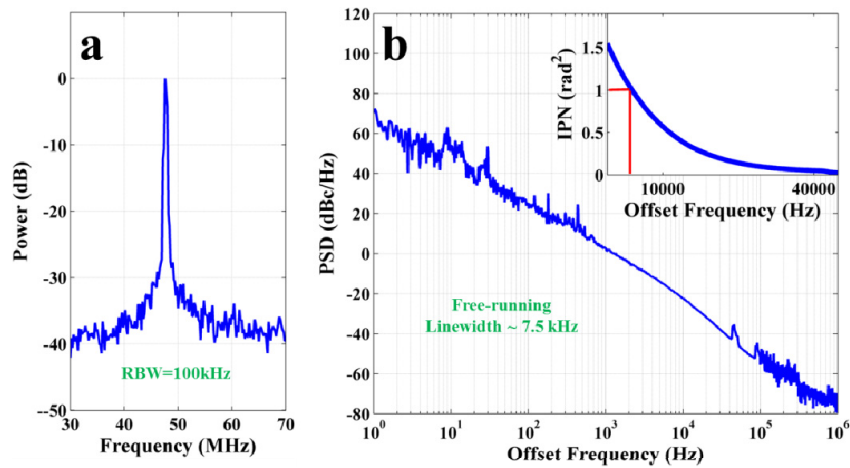


Fig. 4. Free running f_{ceo} measurements. (a): RF spectrum demonstrating $\sim 40\ \text{dB}$ SNR. RBW: resolution bandwidth. (b): power spectral density. Inset: integrated phase noise (IPN). The curve intersects $1\ \text{rad}^2$ at $\sim 7.5\ \text{kHz}$.

3. Carrier envelop frequency f_{ceo} stabilization

For f_{ceo} stabilization, the output of the $f\text{-}2f$ interferometer is sent through narrow bandpass optical filter onto a photodetector and an error signal is generated through RF signal processing (band pass filtering and amplification) and mixing with an RF reference at 60 MHz with a digital phase detector without pre-scaler. This error signal is fed to the oscillator through a low pass filter at 1.9 MHz and an analog PID controller with a servo bandwidth of 200 kHz. Tight phase-locking of the f_{ceo} to an RF reference is permitted by the ultra-low noise property of the laser oscillator. The electrical spectrum of the locked f_{ceo} is shown in Fig. 5(a), in which a servo bump at around 200 kHz is observed. The in-loop f_{ceo} frequency is divided by a factor of 6 for further analysis.

For further characterization of the residual phase error for f_{ceo} stabilization, the power spectral density (PSD) of phase noise and fractional frequency stability in terms of Allan deviation are evaluated. The in-loop RF signal and the RF reference utilized to stabilize the

f_{ceo} are compared to recover the phase error of the stabilization scheme. The measured PSD of the phase noise of the locked f_{ceo} is represented on Fig. 5(b) in blue. The noise level is at -90 dBc/Hz at 1 Hz with a servo bump at 200 kHz, with a white phase noise floor at -110 dBc/Hz. The integrated phase noise from 2 MHz to 1 Hz is ~ 470 mrad, as shown in Fig. 5(b) in green. For applications such as in optical clock metrology, the fractional frequency stability is evaluated in terms of Allan deviation with a very low measurement bandwidth (here 0.5 Hz Nyquist equivalent bandwidth), as optical atomic clocks are limited by white frequency noise, not white phase noise. The residual instability of the phase error of the stabilization scheme exhibits an Allan deviation of 1.83×10^{-4} , corresponding to a frequency resolution of approximately 200 μHz at 1 second averaging time. Scaled to an optical wavelength of 656 nm, the fractional frequency stability is at the level of 4×10^{-19} at 1 second averaging time, as plotted in Fig. 5(c). The phase error averages down to 2×10^{-20} , limited by the spectral purity of the RF reference used to stabilize the f_{ceo} . Moreover, validation has been made to exclude any systematic frequency offset added by the stabilization setup at the level of 1×10^{-20} .

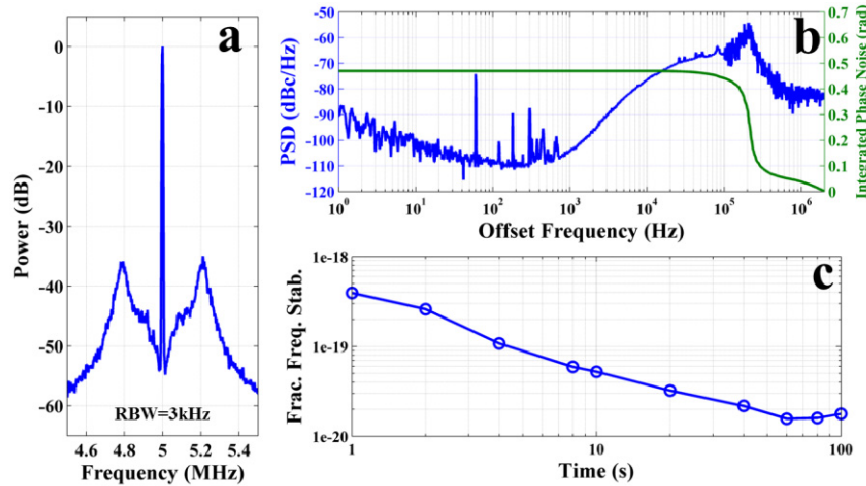


Fig. 5. (a) Electrical spectrum of the stabilized carrier envelope offset frequency in a 3 kHz resolution bandwidth. (b) Blue: power spectral density of the residual phase noise error of the stabilized f_{ceo} ; Green: integrated phase noise from 2 MHz to 1 Hz. (c) Fractional frequency stability of the residual phase noise error scaled to an optical frequency.

In order to stabilize the second degree of freedom of the frequency comb (repetition rate) an actuator such as an electro-optic modulator or a piezoelectric transducer can be readily included in the laser cavity to control the optical path length. In this study, this stabilization is not performed due to the absence of a CW laser whose wavelength falls within the optical spectrum generated.

4. Conclusion

In conclusion, we have demonstrated a self-starting optical frequency comb based on an all polarization-maintaining passively mode-locked Yb-fiber laser oscillator utilizing a nonlinear amplifying loop mirror. The laser oscillator spans from $\sim 1015 - 1070$ nm at ~ 131 MHz repetition rate with a root-mean-square RIN of $\sim 0.18\%$ integrated from 1 MHz to 1 Hz. After super-continuum generation, an optical spectrum spanning from ~ 650 nm to 1400 nm is obtained. Such a spectral range is extremely interesting for the realization of a single-path optical frequency comb to bridge the ultra-high stability offered in the infra-red domain to ultra-high accuracy in the visible domain. The detection of the carrier envelope offset frequency f_{ceo} is realized with an f - $2f$ interferometer. The free running f_{ceo} exhibits a signal-to-

noise ratio of ~ 40 dB in a 100 kHz resolution bandwidth and a linewidth of ~ 7.5 kHz. Thanks to the low free-running noise of the laser oscillator, a servo-bandwidth of 200 kHz is sufficient for f_{ceo} stabilization. The residual phase noise power spectral density, measured in-loop, for f_{ceo} stabilization is -90 dBc/Hz at 1 Hz Fourier frequency with a servo bump at 200 kHz. The integrated phase noise is approximately 470 mrad from 2 MHz to 1 Hz. Experimental measurement of the phase stability reveals an achieved Allan deviation (0.5 Hz measurement bandwidth) at 4×10^{-19} level at 1 second averaging time. Further studies will be devoted to the integration of this optical frequency comb into a full common path metrological system aimed at the realization of optical clocks with 10^{-19} stability in 1 sec as well as exploiting the present system for applications in precision timing distribution.